

ENERGY CHANGES – CELLS & BATTERIES

KEY VOCABULARY

Cell: two different electrodes in an electrolyte producing a voltage.

Battery: two or more cells connected together.

Electrode: a metal that reacts to release or accept electrons.

Rechargeable: a cell whose reaction can be reversed by electricity.

COMMON MISCONCEPTIONS

~~— A single cell is a battery.~~

✓ A battery is two or more cells joined.

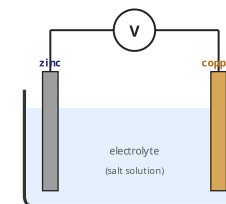
~~— Any two metals give the same voltage.~~

✓ Voltage depends on the difference in reactivity.

~~— All cells can be recharged.~~

✓ Only some — non-rechargeable cells run out.

A SIMPLE CELL



1 MAKE A VOLTAGE

State what is needed to make a simple cell produce a voltage. [2 marks]

2 BIGGER VOLTAGE

Explain how to increase the voltage of a cell using the metals chosen. [2 marks]

3 BATTERY

State the difference between a cell and a battery. [1 mark]

4 RECHARGEABLE

Explain why a rechargeable cell can be used many times. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A bigger difference in metal ____ gives a ____ voltage.

reactivity

larger

electrolyte

voltage

6 WHY IT STOPS

Explain why a non-rechargeable cell eventually stops working. [1 mark]
